Image Restoration with Shared NAFNet and D4 Fine-Tuning

1 Method

The task is to recover clean RGB images from blurred, noisy inputs without knowing the corruption parameters. We adapt the NAFNet architecture and gated restoration blocks of Chen et al. [1], using a residual encoder–decoder with skip connections, widths 32/64/128 and block depths 2/2/7/2/2. To meet the storage budget, our implementation shares convolutions *within each network stage*; normalization and residual gates remain independent at each depth. This gives **286,563 trainable parameters** and **286,942/300,000 stored ONNX tensor elements**. The self-contained opset-18 graph has FP32 input/output $[1, 3, 384, 384]$ and clips outputs to $[0, 1]$.

Only the supplied 1,200 training pairs are used: no external images or pretrained weights. We train on synchronized 256×256 crops with random quarter-turn rotations and horizontal flips. All 120 validation pairs retain the supplied split and full 384×384 resolution. The objective is mean *per-image* log-MSE:

$$m_b = \frac{1}{CHW}\|\hat{y}_b - y_b\|_2^2, \qquad \mathcal{L} = \frac{1}{B}\sum_{b=1}^{B} \ln\big(\max(m_b, 10^{-12})\big).$$

Scratch/native use $\hat{y} = f_\theta(x)$. The D4 stage adapts the geometric self-ensemble used for inference by Lim et al. [2]: $\hat{y} = \frac{1}{8}\sum_{t\in D_4} t^{-1} f_\theta(tx)$. Eight rotated/reflected views share weights and are inverse-transformed before averaging. Here we also *train through this average*, computing the loss after averaging. This training adaptation and within-stage convolution sharing are project choices. Training outputs are unclipped; export uses the same D4 average with final clipping.

2 Training

The submitted checkpoint follows the three-stage schedule below.

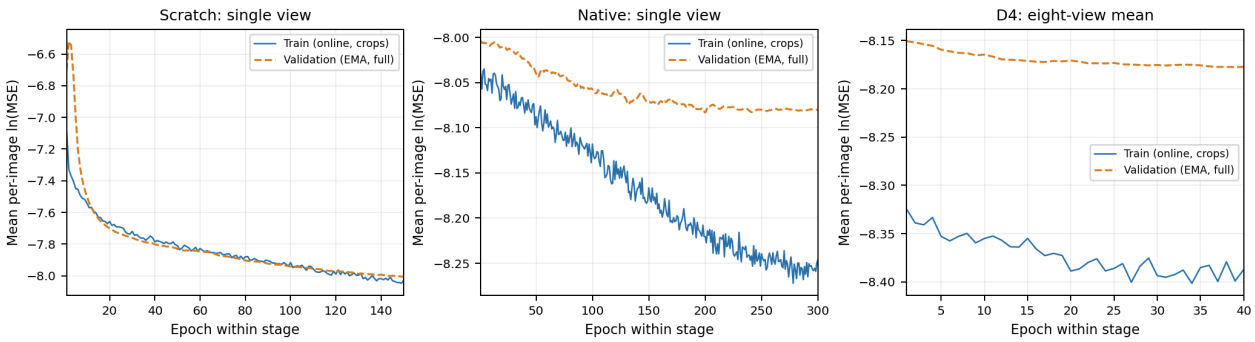
Stage	Epochs	Cosine $T_{\max}$	Initial LR $\to$ $\eta_{\min}$	Batch	Seed	Precision
Scratch	150	300	$10^{-3} \to 10^{-5}$	4	42	FP32
Native	300	300	$5\times10^{-4} \to 2\times10^{-6}$	4	42	FP32
D4	40	40	$7.5\times10^{-5} \to 10^{-6}$	2	4246	BF16

All stages use AdamW ($\beta = (0.9, 0.999)$, $\epsilon = 10^{-8}$, weight decay 10^{-4}), gradient-norm clipping at 1 and EMA decay 0.999. Cosine scheduling steps after each epoch's validation. Scratch was initially planned for 300 epochs, then truncated to 150 as an experimental budget choice, retaining the original cosine schedule. Native loads scratch epoch 150 EMA; D4 loads native **epoch 300 EMA**. Only EMA weights transfer: each stage resets optimizer, scheduler and seeded RNG. D4 also changes the training batch size from 4 to 2. The strict run uses deterministic kernels and seeded shuffling, with TF32 and compilation disabled. BF16 is used only for the D4 forward pass; master weights, loss, EMA and validation remain FP32. We export **D4 epoch 39 EMA**.

3 Results

Official CPU ONNX scoring averages per-image full-image RGB PSNR over all 120 validation images, using float64 scoring and no border crop. The starter score is the reference reported in the assignment README.

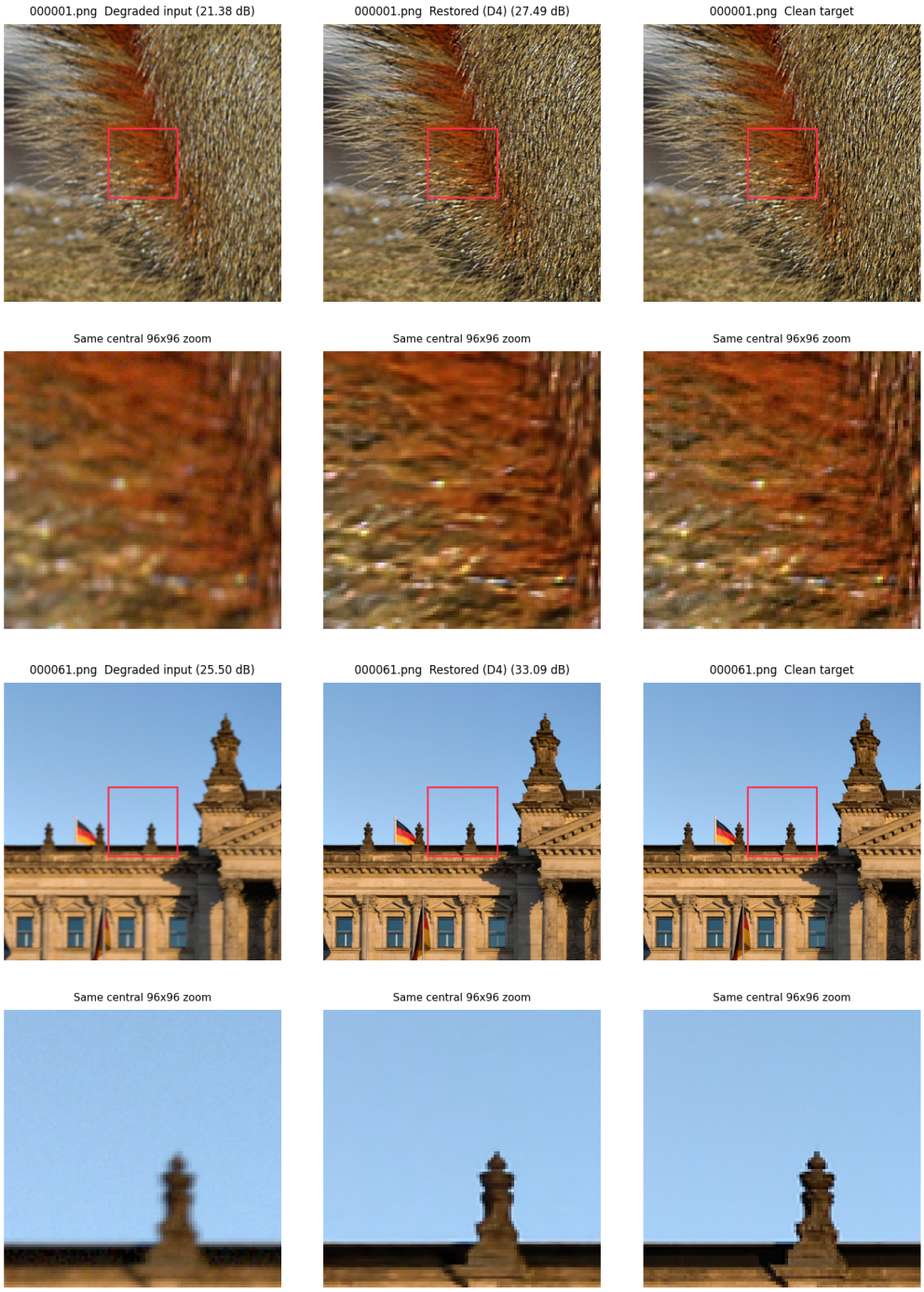
Method	Mean PSNR (dB)
Degraded input	29.7931
Starter CNN	31.74
Shared NAFNet + D4 (ours)	**35.5189**



Training curves. Each panel uses local epochs and its own vertical range, with the same mean per-image ln(MSE) objective: solid training, dashed EMA validation. Stages are not joined. The D4 boundary reflects a new eight-view forward pass, smaller batch and reset optimizer/scheduler, *not missing epochs.*

Qualitative results

Two fixed validation examples show degraded input, our restoration and the clean target. The matching central 96×96 crop is enlarged below each full image; PSNR labels use the full-image metric.



References

[1] Liangyu Chen, Xiaojie Chu, Xiangyu Zhang, and Jian Sun. *Simple Baselines for Image Restoration*. ECCV, 2022. arXiv:2204.04676.

[2] Bee Lim, Sanghyun Son, Heewon Kim, Seungjun Nah, and Kyoung Mu Lee. *Enhanced Deep Residual Networks for Single Image Super-Resolution*. CVPR Workshops, 2017. arXiv:1707.02921.